Roundtable on
Sustainable Biomaterials

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RSB – ROUNDTABLE ON SUSTAINABLE BIOMATERIALS

RSB Proactive Guidance on Co-Processing GHG Calculations

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1. Introduction

Following the publication of the COMMISSION DELEGATED REGULATION (EU) 2023/1640 of 5 June 2023 on the methodology to determine the share of biofuel and biogas for transport, produced from biomass being processed with fossil fuels in a common process in line with the European Union Directive 2018/2001/EU on the promotion of the use of energy from renewable sources (RED II), the RSB-STD-11-001 - Standard for EU Market Access and the RSB-PRO-11-001-20-001 – RSB EU RED Traceability were updated to include the established requirements for Co-Processing.

One of the main issues in Co-Processing is how to assess and allocate greenhouse gas (GHG) emissions to the production process's resulting outcomes.

This document was created to provide further guidance related to greenhouse gas calculations for fuels resulting from biomass being processed with fossil fuels in a common process, expanding information about the requirements stated in the RSB-STD-11-001 - Standard for EU Market Access, in line with the European Union Directive 2018/2001/EU on the promotion of the use of energy from renewable sources (RED II).

2. Scope of this document

This document was created to provide further guidance related to the GHG calculations for fuels resulting from biomass being processed with fossil fuels in a common process and is directly related to the requirements stated in the Standard for EU Market Access [RSB-STD-11-001] and RSB EU RED Procedure for Traceability [RSB-PRO-11-001-20-001].

3. Effective date

Update with publication date.

4. Referenced RSB Standards

- RSB EU RED Traceability [RSB-PRO-11-001-20-001]
- RSB Standard for EU Market Access [RSB-STD-11-001]

5. Version history

Version history of this document is summarised below:

Version	Date	Summary of changes
01	January 2025	-

6. Co-Processing greenhouse gas calculation

6.1. Methodology for GHG Calculation in Co-Processing

The methodology consists of a virtual split of the biogenic conversion pathway, as illustrated in figure 1.

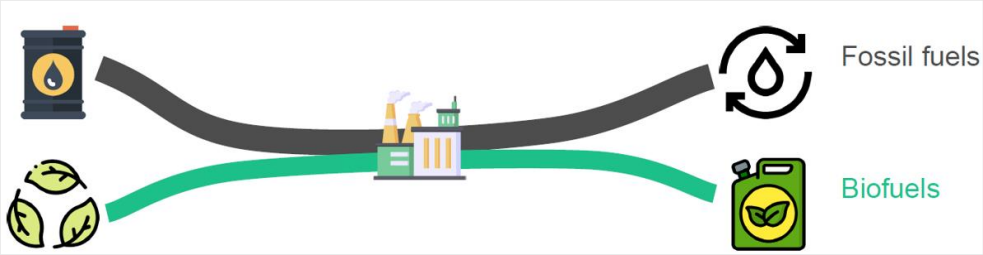


Figure 1: Virtual Split of Biogenic Conversion Pathway. Source: Key aspects of certifying co-processed fuels, of 10 July 2024, European Commission[1].

The methodology includes the following topics:

1. Establish a Baseline Scenario
    - Record inputs and emissions for the refinery operating with only fossil feedstock (fossil-only baseline).
    - Capture energy inputs (e.g., hydrogen, heat, electricity) and operational emissions, including auxiliary emissions (e.g., for pumps).
2. Introduce the Biogenic Feedstock Scenario
    - Introduce biogenic feedstock either as an additional input or as a replacement for a portion of fossil feedstock.
    - Record all new or increased inputs necessary for processing biogenic feedstock, such as extra hydrogen for deoxygenation and additional heat/electricity requirements.
3. Attribute GHG Emissions to Biogenic and Fossil Portions
    - Attribute emissions for energy inputs and reagents required for biogenic feedstock (e.g., hydrogen) directly to biofuel.
    - Allocate refinery-wide energy inputs proportionally between bio-based and fossil-based feedstocks, based on energy content at the inlet.
4. Adjust for Counterfactual Scenario
    - Use the fossil-only baseline as a counterfactual to measure the increase in GHG emissions due to the introduction of biogenic feedstock.
    - Attribute any increases in inputs, compared to the fossil-only baseline, directly to biofuel.
5. Incorporate Transport and Distribution Emissions
    - Apply transport and distribution emissions specifically to the co-processed biofuel according to RED II guidelines.
    - Use disaggregated default values from RED II for transport emissions if specific data isn't available.

[1] More information can be found at: https://energy.ec.europa.eu/topics/renewable-energy/bioenergy/biofuels_en

6. Include Biogenic CO_2 Emissions and Fuel Use Emissions
    - Biogenic CO_2: Account for all upstream emissions associated with biogenic feedstock (e.g., cultivation, land use, transport). Exclude CO_2 from direct biogenic material if the feedstock meets sustainability criteria.
    - Fuel Use Emissions: Include emissions from the combustion of the biofuel in its final use. Use RED II Annex V values for default fuel use emissions or measured data if available.

Considering the methodology, Participant Operators can use the following equation to calculate the GHG emissions of a specific co-processed liquid fuel $E_{biofuel}$, in g CO_2eq/MJ:

$$E_{biofuel} = e_{upstream} + e_{processing} + e_{transport} + e_{fuel\ in\ use}$$

Where:

$e_{upstream}$: Upstream emissions from biogenic feedstock production, including cultivation, land use, and transport to the refinery.

$e_{processing}$: Processing emissions, calculated as:

$$e_{processing} = \frac{E_{total\ process} \times F_{bio}}{F_{bio} + F_{fossil}} + e_{additional}$$

Where:

$E_{total\ process}$: Total emissions from the baseline process shared between fossil and biogenic inputs.

F_{bio}: Energy fraction of the biogenic feedstock at the inlet.

F_{fossil}: Energy fraction of the fossil feedstock at the inlet.

$e_{additional}$: Additional emissions from energy or reagents (e.g., extra hydrogen or heat, including upstream emissions of those additional inputs) specifically required for biogenic feedstock.

$e_{transport}$: Emissions from transport and distribution of the final biofuel.

$e_{fuel\ in\ use}$: Emissions from the combustion of the biofuel during its final use, based on default or measured values as outlined in RED II Annex V.